

## APPLIED PHYSICS

**I Semester:** CSE, IT, CSE-Data Science, CSE-Cyber Security, CSE-AIML, CSIT, ECE, MECH, AERO, EEE

| Course Code: | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|--------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A6BS08       | BSC      | L            | T | P | C       | CIA           | SEE | Total |
|              |          | 3            | 1 | 0 | 4       | 40            | 60  | 100   |

### Course Objectives:

1. Learn the basic principles of quantum physics and its applications
2. Understand the formation of energy bands and atomic structure in solids for material classification
3. Understand the underlying mechanism involved in construction and working properties of different types of semiconductor devices
4. Learn the basic principles of laser and optical fiber in information technology
5. Identify the importance of nanoscale and various fabrication and characterization techniques and quantum computations in engineering applications

### UNIT-I Quantum Physics

Black body radiation, Stefan-Boltzmann's law, Planck's radiation law (Qualitative treatment), Photoelectric effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation, Physical Significance of the wave Function, Particle in One Dimensional Potential Box.

### UNIT-II Electronic properties of Materials & Band theory of solids

**Electronic properties of Materials:** Classical free electron theory and Quantum free electron theories of metals, success and drawbacks, Bloch theorem, Kronig-Penny model (Qualitative treatment), E-k diagram, effective mass of electron.

**Band theory of solids:** Origin of energy band formation in solids, Fermi energy level, Fermi-Dirac distribution law, Classification of materials as conductors, insulators and semiconductors.

### UNIT-III Semi-conductors & Semiconductor Devices

**Semiconductors:** Intrinsic and Extrinsic Semiconductors, formation of PN junction diode and its V-I characteristics, Direct and Indirect band gap semiconductors, Hall effect and its applications.

**Semiconductor Devices:** Construction, working and V-I characteristics of PIN Diode, LED, Solar cell and their applications.

### UNIT-IV Laser & Fiber Optics

**Laser:** Characteristics of Laser, Absorption, Spontaneous and Stimulated emission of radiations. Lasing actions-Pumping mechanism, Meta stable state and Population inversion, Nd-YAG laser, CO<sub>2</sub> laser, Applications of lasers in different fields.

**Fiber Optics:** Structure of fibers, Total Internal Reflection, Acceptance angle – Numerical Aperture, Types of fibers- SI and GI fibers, Single and Multimode fibers - SMSI, MMSI, MMGI, Fiber Optic Communication system, Signal Degradation - Attenuation mechanism, Dispersion, Applications of fibers in different fields.

### UNIT-V Physics of Quantum computing & Quantum gates

**Physics of Quantum computing:** Idea of classical bits and qubits, advantages with qubits over classical bits, Bloch vector representation of state of qubits.

**Quantum gates**-Single qubit logic gates- Pauli X, Y, Z and Hadmard gate in matrix form, Two level gates-CNOT and SWAP gates and representation in matrix form, Comments on No cloning theorem, Entanglement, Quantum Teleportation – Basic Idea, Quantum Key distribution protocol - BB84 protocol.

### Text Books:

1. P.K Palanisamy, Engineering Physics, Sitech Publications, 2013, IVth Ed.
2. Nielsen M. A., I. L Chung, Quantum Computation & Quantum Information, Cambridge Univ. Press.
3. M. N. Avadhanulu, P.G. Kshirsagar & TVS Arun Murthy" A Text book of Engineering Physics"- S. Chand Publications, 11th Edition 2019.

**Reference Books:**

1. Quantum Physics, H.C. Verma, TBS Publication, 2<sup>nd</sup> Edition 2012
2. Fundamentals of Physics – Halliday, Resnick and Walker, John Wiley & Sons, 11<sup>th</sup> Edition, 2018.
3. Engineering Physics by Shatendra Sharma and Jyotsna Sharma, Pearson Publication, 2019
4. B.K Pandey and Chaturvedi, Engineering Physics, Cengage Learning, 2<sup>nd</sup> Edition, 2022

**Course Outcomes : Students should be able to:**

At the end of the course, students will be able to:

1. Analyze the microscopic properties of materials using principles of quantum physics for engineering applications
2. Explain the behavior of different electronic materials based on the concepts of band theory
3. Apply the knowledge of Solar PV cells for choice of materials in efficient alternate energy generation
4. Gain the knowledge of production of laser and usage of fibers in fiber optic communication technology
5. Comprehend the knowledge of quantum physics in quantum computation for secure information technology